Prediction and Classification of Students' Performance

Not only is education important for the personal development of an individual, but it also plays a big role in the economic growth rate of a country. Countries with a more educated workforce, could imply a greater enhancement in skills and workplace productivity, leading to an increase in overall economic growth. As such, the prediction and classification of students' performances as early as their schooling years is a vital area of research. Through the early prediction of a student's grades, early intervention strategies can be implemented by the school authorities with the help of parents. By determining students who are at risk of performing poorly, they could inform the respective parents for further action, besides providing extra tutoring or counselling if necessary, to aid in enhancing their grades in the next period. Moreover, the classification of students' grades can help the authorities to organise the data well and track the academic progress of their students over time. Classifying students according to their grades will also help in identifying specific patterns in each group to enable more targeted approaches in helping them to improve themselves, besides providing relative comparisons across groups for the review of parents.

The most important features in predicting and classifying students' grades are also a very important aspect to take into consideration. By identifying features that have the most impact on a student's good or bad achievements, the school authorities can allocate their resources efficiently to help improve that aspect of the student's life if possible. This would provide a more targeted intervention instead of blindly carrying out programs that focus on irrelevant aspects, which not only wastes time, but also energy and money.

Data Set

In building models to predict and classify students' grades, the Student Performance data set by Cortez (2014) is utilised. This data set consists of student achievements in the Mathematics and Portuguese subjects, with the students coming from two different secondary Portuguese schools, namely Gabriel Pereira and Mousinho da Silveira. Data collection was done via school reports and questionnaires, in which the data comprises of 649 instances and 33 attributes, including student grades, demographic, social and school related features. The student grades are commonly used as the target variables, and it is important to note that in the Portugal education system, student performances are graded on a 20-point system, where 0 represents the lowest grade and 20 represents the highest. In this data set, the variables G1, G2 and G3 represent the first period, second period and final grades respectively, as three periods are used to evaluate students throughout the academic year. For this short analysis, only the mathematics data set is utilised, where it contains 395 instances.

Decision Trees: General Procedure and Maximum Depth Selection

Regression and classification trees are examples of machine learning techniques for creating prediction models from data (Loh 2011). The models are constructed by fitting a simple prediction model within each of the recursively partitioned data spaces, enabling the partitions to be graphically depicted as a decision tree (Breiman et al. 1984). These decision trees are hierarchically arranged branching structures that express a collection of rules, which can be rewritten as a collection of IF-THEN rules. This is what makes decision trees so popular, they are easily interpretable and explained since they more closely resemble the way humans make judgements. As such, the analysis utilises decision trees in predicting student performance as it provides a transparent portrayal of the decision-making process that is essential for school authorities and parents in understanding the factors that strongly affect their students and children. Not to mention, decisions trees have a swift training process as they can handle a large number of features without being too computationally extensive.

Regression Tree

In predicting the students' first period grades (G1), a regression tree is built using Python. After reading in the data, it is checked for any null values. Apparently, the data is already clean as there are no missing values for any of the variables available. As such, dummy variables are then created for all the categorical variables that are not already in a numeric format. It is important to note that G1, G2 and G3 are all very highly correlated with one another with values of 0.8521 (G1 and G2), 0.8015 (G1 and G3) and 0.9049 (G2 and G3). Following suit, the G2 and G3 variables are dropped from the list of independent variables as to avoid any multicollinearity issues that may persist. G1 is also dropped since it is the target variable. Besides, it would make for a more useful prediction model if those grades are not included, since the inclusion of grades would be sure to make these the most important variables in determining the target variable. By excluding all the period grades from the independent variables list, the most important variables from the list of demographic, social and school related features can be identified. This is much more useful information to the school authorities as they can pin-point the factors most influencing their students' performances and thus take the right reactionary measures.

The data is then split into training and testing sets following the 80:20 rule with a random state of 1. Maximum depth values of 2 to 15 are tested for the regression tree through a for loop and by looking at their test MSE scores. Figure 1 below shows that a *max_depth* value of 2 gives the lowest MSE score of 7.459. The results for the fitting of the regression tree on the training set with a *max_depth* of 2 are discussed in the following section.

```
Max Depth: 2, Test MSE: 7.4594640215069905
Max Depth: 3, Test MSE: 8.237563639801373
Max Depth: 4, Test MSE: 7.639956691928665
Max Depth: 5, Test MSE: 10.84999101779639
Max Depth: 6, Test MSE: 10.207435383208532
Max Depth: 7, Test MSE: 9.283211923701815
Max Depth: 8, Test MSE: 10.543883543588793
Max Depth: 9, Test MSE: 9.717708781769456
Max Depth: 10, Test MSE: 10.212896963271874
Max Depth: 11, Test MSE: 14.986833864347428
Max Depth: 12, Test MSE: 11.091954131863716
Max Depth: 13, Test MSE: 11.366912798874823
Max Depth: 14, Test MSE: 12.71696993670886
Max Depth: 15, Test MSE: 12.963123223973135
```

Figure 1 Test MSE values for regression trees with *max_depth* values of 2 to 15

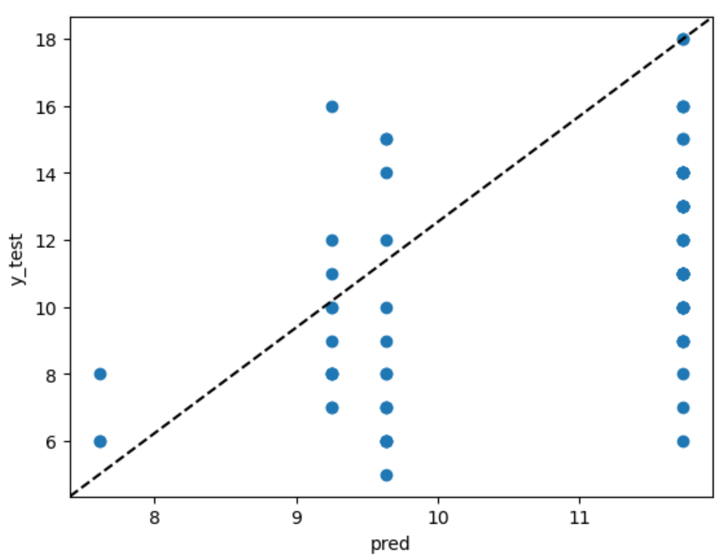

Figure 2 Plot of actual versus predicted values by the regression tree model from the test set

Figure 2 on the other hand shows the plot of the actual versus predicted values in the test set that are generated by the regression tree. It seems that only four prediction values are made by the model since only a *max_depth* of 2 is used. Since the MSE value is 7.459, the square root of it would give an RMSE value of 2.731. The RMSE is essentially a measure of the average difference between the predicted values and the actual values. This indicates that on average, the predicted test values from the regression tree model are within approximately 2.731 points of the true first period grade of students.

Classification Tree

In classifying the students' second period grades, G2, a classification tree is built. Similar as was previously done, the data is read in and dummy variables are generated. Next, G2 is transformed into G2T with five categories, namely A (17 to 20), B (13 to 16), C (9 to 12), D (5 to 8) and E (0 to 4), for a five-class classification model. The variables G1, G2, G3 and G2T are dropped from the independent variables list for the same reasons as before, G2 is also dropped since it is now called G2T with five different grades instead of a continuous numeric variable. G2T is the target variable. Once again, the 80:20 rule is used for the splitting of the training and testing sets. To obtain the best *max_depth* value in building the classification tree, the training and testing accuracies are tested for values 2 to 15 as shown in Figure 3. A classification tree with a *max_depth* value of 4 gives the best testing accuracy of 0.5823. This increase from the training accuracy of 0.5570 shows that the model is not overfitting the training data. This could be an indication that the model has slightly improved as it learns to detect underlying patterns in the training data and generalise to new and unseen data from the testing set. The classification tree discussion along with the most important variables in classifying G2T can be found in the next section.

```
Max Depth: 2, Training Accuracy: 0.4430379746835443, Test Accuracy: 0.3037974683544304
Max Depth: 3, Training Accuracy: 0.5063291139240507, Test Accuracy: 0.5063291139240507
Max Depth: 4, Training Accuracy: 0.5569620253164557, Test Accuracy: 0.5822784810126582
Max Depth: 5, Training Accuracy: 0.5949367088607594, Test Accuracy: 0.4810126582278481
Max Depth: 6, Training Accuracy: 0.6582278481012658, Test Accuracy: 0.3924050632911392
Max Depth: 7, Training Accuracy: 0.731012658227848l, Test Accuracy: 0.4430379746835443
Max Depth: 8, Training Accuracy: 0.8006329113924051, Test Accuracy: 0.4177215189873418
Max Depth: 9, Training Accuracy: 0.870253164556962, Test Accuracy: 0.3924050632911392
Max Depth: 10, Training Accuracy: 0.9177215189873418, Test Accuracy: 0.4050632911392405
Max Depth: 11, Training Accuracy: 0.9525316455696202, Test Accuracy: 0.35443037974683544
Max Depth: 12, Training Accuracy: 0.9683544303797469, Test Accuracy: 0.379746835443038
Max Depth: 13, Training Accuracy: 0.990506329113924, Test Accuracy: 0.3670886075949367
Max Depth: 14, Training Accuracy: 1.0, Test Accuracy: 0.379746835443038
Max Depth: 15, Training Accuracy: 1.0, Test Accuracy: 0.3670886075949367
```

Figure 3 Training and testing accuracy values for classification trees with *max_depth* values of 2 to 15

	A	B	C	D	E
A	0	0	0	0	0
B	1	6	9	1	0
C	1	9	34	8	0
D	0	0	2	6	0
E	0	0	0	2	0

Figure 4 Confusion matrix of test data set

The confusion matrix in Figure 4 portrays the actual categories/grades in the columns and the predicted values in the rows. It seems that both of the students who are supposed to be in category A were wrongly classified, 9 out of 15 in category B were wrongly classified, 11 out of 45 in category C wrongly classified and 11 out of 17 wrongly classified for D. There are no E categories in the test set. Besides category A with only two instances, the model is very bad at classifying instances in category D. It does a satisfactory job in classifying instances of category C. This could be due to the fact that there are many instances for category C but less for the others. Maybe in further research, generating synthetic samples could be done for a more equal class distribution to enable proper learning of the characteristics in each category.

In a paper by Cortez and Silva (2008), they have built prediction models to predict the final grade, G3 and found that the predictive accuracy is good when G1 and/or G2 are included. In their five-level classification model using a decision tree on the mathematics data set, it was found that the Percentage of Correct Classifications (PCC) value is only 31.5%. The values by including only G1 or including both G1 and G2 on the other hand are 57.5% and 76.7% respectively. This further supports the bad testing accuracy of 58.23% that was achieved by our analysis, indicating that decision trees might not be such a good choice for a five-level classification problem on student grades if there are no previous or current grades present as independent variables. It could be that the model is unable to capture complex relationships in the data.

For the regression tree, the RMSE results of the researchers are 1.94, 2.67 and 4.46 for the no dropping of grades, dropping G2, and dropping G1 and G2 respectively. As such, other more advanced models could be tried out such as random forests, boosting, neural networks, Support Vector Machines and so on to build more accurate prediction and classification models although they would remain almost the same without other grade variables. Nevertheless, these models provide an important view into the most important variables affecting the students' performances.

Regression Tree Results

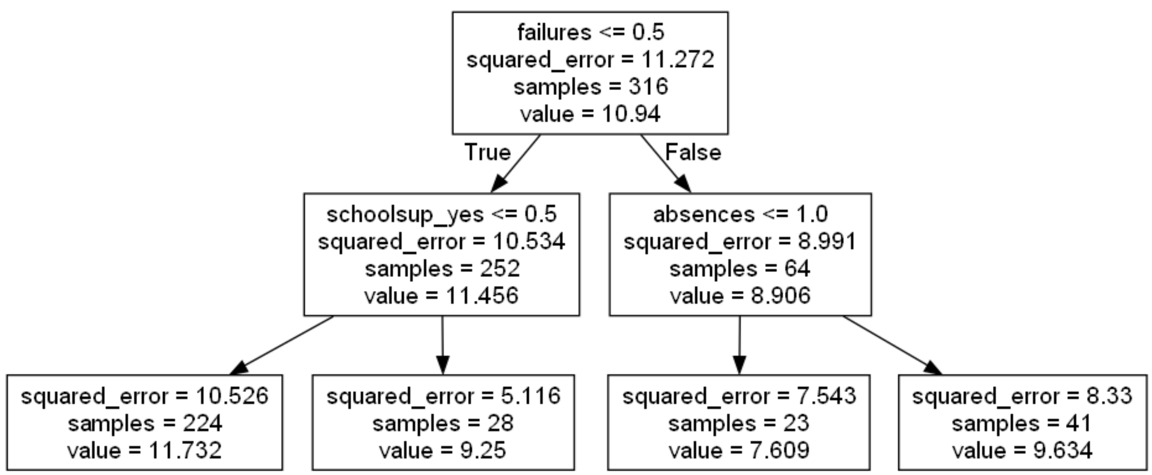

Figure 5 Regression tree for the Student Performance data set

The graphical representation of the regression tree can be found in Figure 5, whereas Figure 6 provides a horizontal bar chart of the variable importance for the regression tree. Here, the regression tree provides predictions of the first period grade of students. The components in the tree are as follows:

- *squared_error*: squared difference between the predicted and actual values in the training set.
- samples: the number of instances in that region
- value: computed from the mean value of the target variable within that region

Based on the regression tree, the *failures* variable is the most important feature to predict the first period grades seeing as it appears at the root of the tree, with three out of 30 features being utilised by the model. The four predicted values by this model, 11.732, 9,25, 7.609 and 9.634, minimises the squared error. The variable *failures* is simply the number of past class failures that a student has, *schoolsup* is whether the student received extra educational support or not, whereas *absences* is the number of school absences a student has.

Out of 395 instances, 316 are used in the training set. These 316 instances have a squared error of 11.272 and a mean G1 value of 10.94. The tree first divides students based on whether they have faced any past class failures before or not. Turns out, 252 students have never had failures from previous classes while 64 of them have had setbacks. *schoolsup* then played a role in determining the scores of students with no failures, where they were divided once again, 224 of them have not received extra educational support while 28 have received extra educational support. This is because if *schoolsup_yes <= 0.5*, it indicates a value of 0, which means no extra educational support. On the other hand, *absences* plays a role in determining the grades of students who have had at least one past class failure before. Out of the 64 students with failures, their mean grade is 8.906, 2.55 points lower than students with no past failures. These students with past failures are then divided into two groups, 23 of them who have had 0 or 1 school absences before, and 41 of them with more than 1 absence. Apparently, students with no failures and no extra educational support are predicted to have the highest G1 grades (11.732), followed by students with failures and more than one absences (9.634), students with no failures and receive extra educational support (9.25), and the worst grade goes to students with failures and 0 or 1 absences (7.609). It is quite logical for those with no past class failures before to stay consistent as they already have the capacity and capabilities to excel in their studies. As a matter of fact, those with no past failures or extra educational support might be a group of really excellent students, and as such, they do not need more support as they are already capable of studying on their own pace. Moreover, weaker students might already be facing academic difficulties, indicating their need to receive extra educational support to make them understand the syllabus well and able to achieve better outcomes or maintain their grades as is.

It is quite intriguing however, that students with more absences are predicted to score better than students with less absences. It could be that these students with minimal absences struggle in their mathematics subject or they are disinterested in the subject, leading to lower levels of class engagement even though they attend classes regularly. Merely attending school every day does not guarantee better results if these students do not work on themselves. It is also important to bear in mind that the other two variables, *failures* and *schoolsup* could be more important in determining the G1 scores and thus outweigh the contribution of the *absences* factor. This will be explored next.

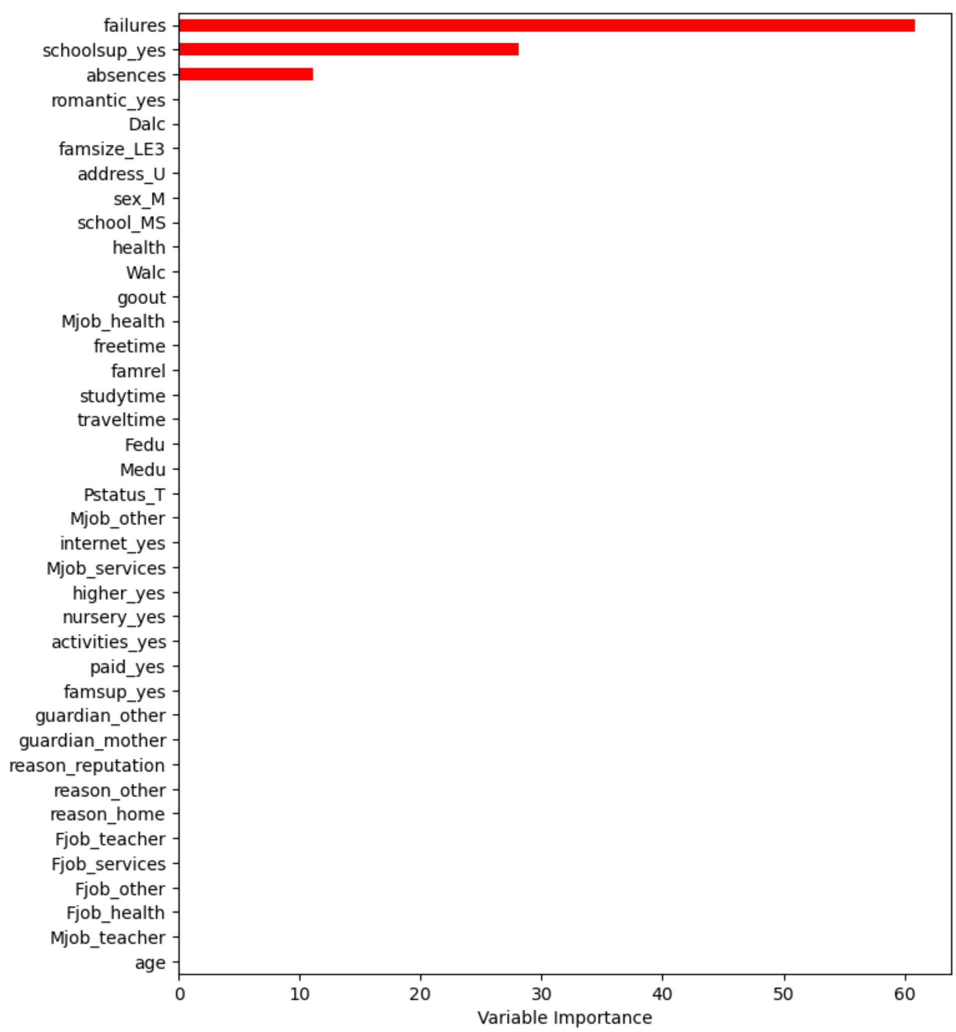

Figure 6 Variable importance plot for the regression tree

This plot provides us with an interpretation of the contribution of each variable in the predictive model. The results of this chart confirms our initial assumptions in that ***failures*** **is the most important variable in predicting G1, contributing 60% of the importance, followed by** ***schoolsup*** **with a 30% importance and lastly** ***absences*** **with a 10% importance.**

As such, the school authorities should take into consideration an implementation of strategies to improve their students' performances based on the most influential factors. For example, they could separate students who have at least one past class failure before from those who have had none. Therefore, they can provide additional mentoring to these students as they have already been identified earlier as weaker students. Furthermore, extra educational support can be provided to all students no matter they have had failures before or not, except for the group of really bright students that might prefer to go on their own pace, as they are capable of looking out for themselves. As for the students with minimal absences, the school could monitor them more closely to determine what is affecting their lower grades, get them to be more engaging and provide help accordingly.

Classification Tree Results

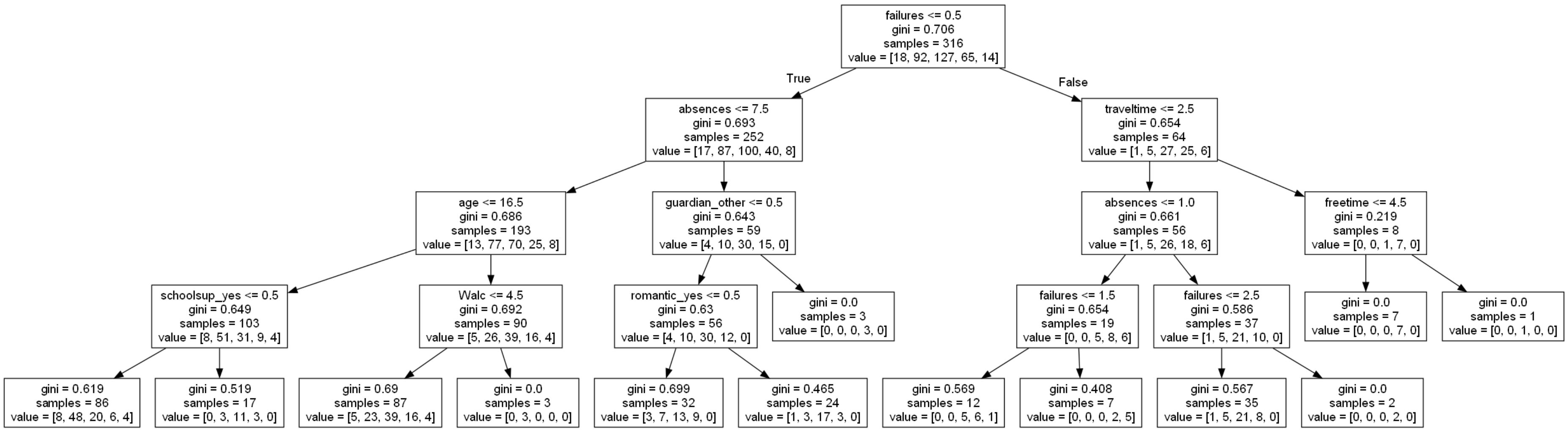

Figure 7 Classification tree for the Student Performance data set

Figure 7 portrays a graphical representation of the classification tree, while Figure 8 provides a horizontal bar chart of the variable importance for the classification tree. The classification tree is used to classify and predict the categories for students' G2T grades based on the various independent variables. For the classification tree, the Gini index is used to measure the impurity of each node. By definition, the Gini index is a measure of node purity, where a low number signifies that most of the observations in a node come from a single class. The samples represent the number of instances remaining to classify and value displays the number of samples in category A, B, C, D and E respectively, following the order given. According to the classification tree, the most important indicator in classifying the second period grades, seem to be *failures* once again as it appears in the root node. Here, there are 9 out of 30 variables deemed as important in the classification model as explained below:

- *failures*: number of past class failures
- *absences*: number of school absences
- *traveltime*: home to school travel time (1 for less than 15 minutes, 2 for 15 to 30 minutes, 3 for 30 minutes to 1 hour, or 4 for more than 1 hour)
- *age*: student's age
- *guardian*: student's guardian (either mother, father or other)
- *freetime*: free time after school (category 1 to 5 indicating very low to very high)
- *schoolsup*: extra educational support
- *Walc*: weekend alcohol consumption (category 1 to 5 indicating very low to very high)
- *romantic*: student has a romantic relationship or not

In the root node, all 316 samples in the training set are present, with a Gini index value of 0.706, being split based on *failures*. This high value indicates that the samples are distributed across multiple categories, as can be seen in value where 18 samples belong to category A, while 92, 127, 65 and 14 belong to categories B, C, D and E respectively. The splits along the tree should keep on creating nodes with lower Gini index values for purer nodes, and dominant class for better classification. The 316 instances are split into two parts based on whether the students have had any past class failures or not. There are 252 students with no past class failures, with 17, 87, 100, 40 and 8 belonging to categories A, B, C, D and E respectively. The students are then split according to *absences*, where 193 of them have 7 or less absences and 59 of them are absent to school 8 or more times. From the 59 who are frequently absent, 3 of them have 'other' as a guardian and will be classified in category D. The other 56 students have their mother or father as their guardian, while being split further according to *romantic_yes*. Out of these 56 students, 32 are not in a romantic relationship while 24 are in a romantic relationship. Since observations will belong to the category that most commonly occurs in the region where it is placed, those students with or without a romantic relationship will be categorised in C. Although, students with a romantic relationship have a lower Gini index value, indicating that category C for these students is more dominant and apparent.

All in all, there are 13 pathways that can be taken to be classified into 5 different categories. To sum up the tree, some points are presented below:

1. Students with no failures, 7 or less absences, aged 15 to 16, and do not obtain extra educational support, are classified to be in category B.
2. Students with no failures, 7 or less absences, aged 15 to 16, and obtain extra educational support, are classified to be in category C.

- This is similar as before in that students who already excel might not need the extra support, whereas those who are weaker in their studies need the support to maintain their grades and not worsen over time.

3. Students with no failures, 7 or less absences, aged 17 to 22, and have a weekend alcohol consumption from levels 1 to 4, are classified to be in category C.
4. Students with no failures, 7 or less absences, aged 17 to 22, and have a very high weekend alcohol consumption of level 5, are classified to be in category B.

- This comes as quite a surprise as those who are heavier drinkers score better. The school authorities could investigate this further as decision trees might not imply a direct causal relationship. As such, they could investigate the relationship between alcohol consumption and second period grades further to gain more insights before making any decisions.

5. Students with no failures, 8 or more absences, father or mother as a guardian, and do not have a romantic relationship, are classified to be in category C.
6. Students with no failures, 8 or more absences, father or mother as a guardian, and have a romantic relationship, are classified to be in category C as well, although the dominance of category C here is higher due to the lower Gini index. The variability between instances here is less when predicting the outcome.
7. Students with no failures, 8 or more absences and have 'other' as a guardian, are classified to be in category D.

- This indicates that students taken care of by their parents tend to perform better. The school could look into the home situation of the students taken care of by 'other' and maybe provide them with extra support in the form of motivation, counselling, tutoring and so on. However, it is important to note that only three students are taken care of by 'other'. As such, the sample could be insufficient.

- The romantic relationship of students does slightly impact their grades following the fact that those with a romantic relationship are more likely to be classified as having a C grade than those without a romantic relationship. The school might need to provide these students with solid advice, especially on how to divide their time well for studies and personal reasons so that they do not get too carried away in their relationship.

8. Students with at least one failure, home to school travel time up to 30 minutes, 0 or 1 school absences, and less than 2 failures (meaning only one failure), are classified to be in category D.
9. Students with at least one failure, home to school travel time up to 30 minutes, 0 or 1 school absences, and at least 2 failures (meaning 2 or more failures), are classified to be in category E.
10. Students with at least one failure, home to school travel time up to 30 minutes, more than 1 school absences, and less than 3 failures (meaning one or two failures), are classified to be in category C.
11. Students with at least one failure, home to school travel time up to 30 minutes, more than 1 school absences, and at least 3 failures (meaning three or more failures), are classified to be in category D.

- This shows the high influence of past class failures on grades. Those students with more failures can be identified quickly for extra aid.
- Also worth noting, it is similar to the regression tree where students with more absences are categorised to have better grades.

12. Students with at least one failure, home to school travel time more than 30 minutes, have free time after school from levels 1 to 4, are classified to be in category D.
13. Students with at least one failure, home to school travel time more than 30 minutes, have a lot of free time after school with level 5, are classified to be in category C.

- This could be because students who have more free time are better at managing their time, so they can allocate enough time for studies. It could also indicate that they have more time for their personal life, reducing stress levels and thus lead to better grades. As such, the school authorities could teach and advice the students on how to manage their time well so that they can have more free time and not feel so burdened by their workload.

From the decision tree, it is very clear that *failures* plays a substantial role in classifying second period grades. This is because all those students with no past class failures are most likely to be classified in categories B and C, with the exception of one D when the guardian is ‘other’, whereas those with at least one failure are classified in categories C, D and E. Students with past failures could lack preparation or discipline, leading to further bad results in the current periods. They might also lack the motivation or be discouraged to do better since they feel disappointed and not confident with their academic abilities. As such, the school needs to create targeted interventions as each of these students with past class failures could be doing badly due to numerous reasons that need to be determined.

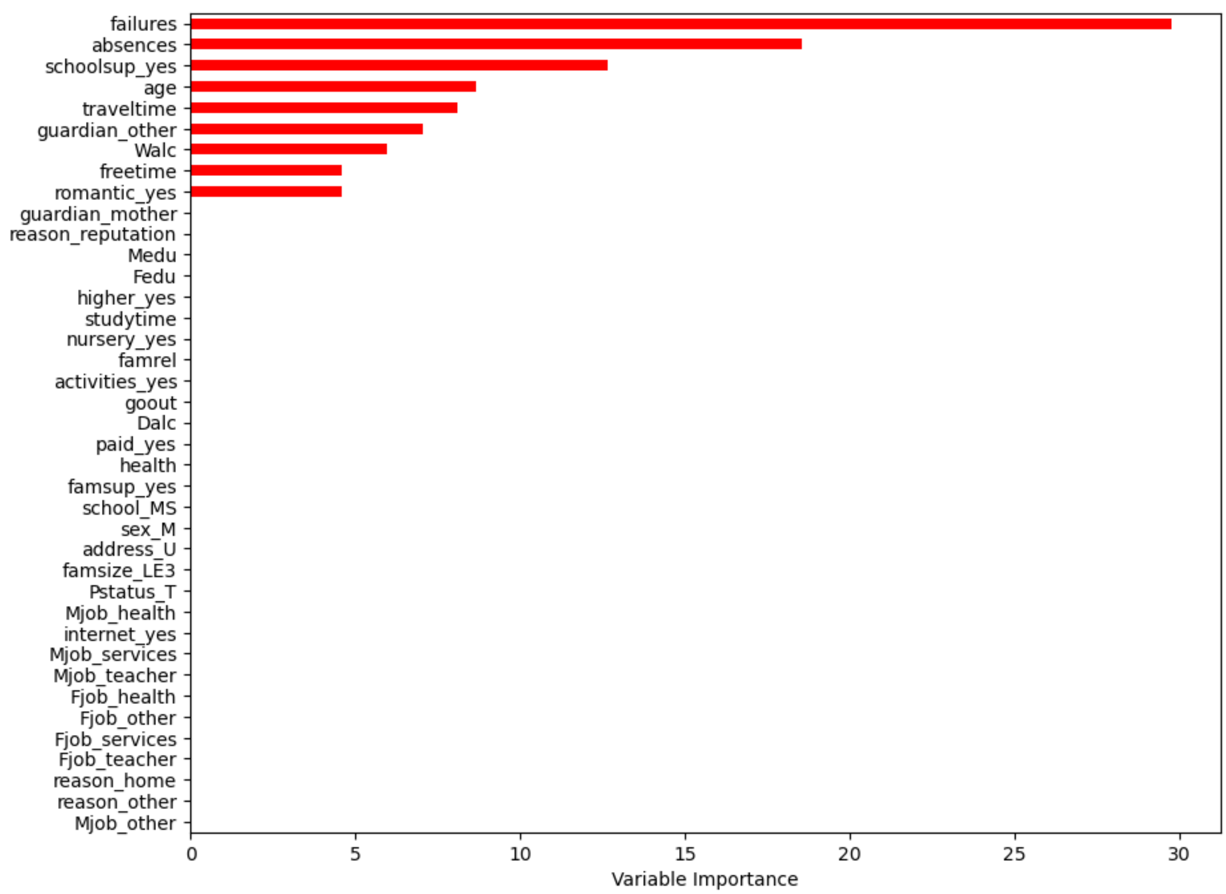

Figure 8 Variable importance plot for the classification tree

For the variable importance plot, the importance is computed using the mean decrease in Gini index. **The plot displays *failures* (30%) as the variable that is most important in classifying the second period grades, followed by *absences*, *schoolsup*, *age*, *traveltime*, *guardian*, *walc*, *freetime* and *romantic*.** It is worth noting that the *failures* contribution to the classification tree is much higher than the others, emphasising its importance, as past performances can be an indicator of the students' academic capabilities and highly affect their achievements. As before, *absences* and *schoolsup* also play really big roles. These three top influential features for the model are all related to school and how students perform or engage. Worth noting, Cortez and Silva (2008) also found that *failures*, *schoolsup* and *absences* were three of the five variables with the highest relative importance for both the regression and classification problems, but by using Random Forests.

The school authorities need to be more alert and look into all these features mentioned for better decision making in enhancing the students' performances. They could focus small first by emphasising only on the most important factors and improve the grades bit by bit before moving on to another strategy for a different feature.

References

Breiman, L., Friedman, J.H., Olshen, R.A. & Stone, C. J. 1984. Classification and regression trees. *Biometrics* 40(3): 874.

Cortez, P. 2014. Student Performance. *UCI Machine Learning Repository*. https://archive.ics.uci.edu/dataset/320/student+performance [22 June 2023].

Cortez, P. & Silva, A. 2008. Using data mining to predict secondary school student performance. *Proceedings of 5th Annual Future Business Technology Conference*, pp. 5–12.

Loh, W.-Y. 2011. Classification and regression trees cross-references. *WIREs Data Mining Knowl Discov* 1(1): 14–23.